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Cryptology assignment - 5

① Encrypt the message 'Enjoy your Journey' using Rail Fence with depth 2 and 3

⇒ Depth 2

E J Y O R O U R N E Y
N O Y U J U R N E Y

cipher: EJYORORE NOYUJUNY

Depth 3

E N J O Y Y U R J U R N E Y
J O U J O U R N E Y

cipher: EVRR NOYUJUNY JOOE.

② Encrypt the message as per your choice using Rail Fence with depth 2 & 3.

⇒ Depth 2

G O O D A Y
O D A Y

plaintext: good day

cipher: G O O Y O O A

Depth 3

G O D A Y
O D A Y

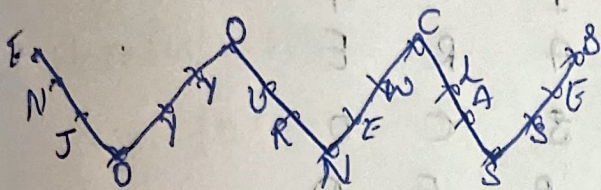
cipher: G O O O A O Y

③ Decrypt the cipher 'EDCSNYUWLEJTYREASONS' encrypted using Rail fence with depth 4

⇒ Depth: 4 length: 19

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

17 18 19



plaintext: ENJOY YOUR NEW CLASSES.

④ Decrypt the cipher 'BOOI6UYUBO8MOUTYREAS' encrypted using Rail fence with depth 3

⇒ Depth: 3 length: 20

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

18 19 20

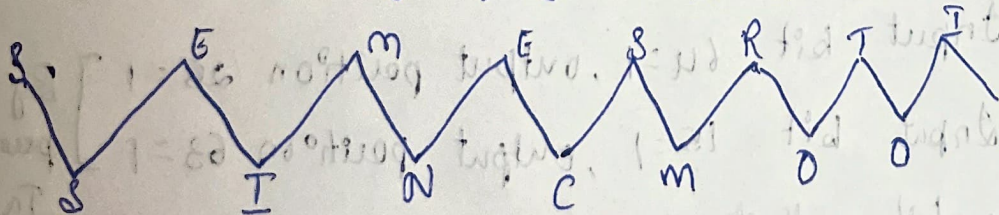
B O I 6 U Y U B O 8 M O U T Y R E A S

plaintext: BUY YOUR BOOKS IN AUGUST

⑤ Decrypt the cipher "SEMESTERISINCMOO" encrypted using Rail fence with the depth 2

⇒ Depth: 2 length: 16

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15



plaintext: SEMESTERISINCMOOT

② Given the 4-bit plaintext (m) 1101 and key (k) 1011, calculate the resulting ciphertext (c) using a bitwise Exclusive-OR (XOR) operation. Show your bit-by-bit work.

→ plaintext: 1101

key: 1011

$$\begin{array}{rcccc} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ \hline 0 & 1 & 1 & 0 \end{array}$$

XOR

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

ciphertext: 0110

③ The input to S-box 1 is 000000. What is the output.

→ 000000

1st

6th bit

1st & 6th bit → row

middle 4 bits → column

So, 0th row & 0th column the number in S-box = 14

Binary conversion of 14 → 1110

output = 1110